4.1절

1

(1) $A = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$

$$\det(A-\lambda I) = \det\begin{pmatrix} 1-\lambda & 0 \\ 2 & 2-\lambda \end{pmatrix}$$

$$= (1-\lambda)(2-\lambda) = 0$$

$\lambda = 1, 2$

① $\lambda = 1$

$$(A-I) = \begin{pmatrix} 0 & 0 \\ 2 & 1 \end{pmatrix}$$

$2x_1 + x_2 = 0$

$2x_1 = -x_2$

$\therefore \begin{pmatrix} k \\ -2k \end{pmatrix}$ 고유 vector

② $\lambda = 2$

$$(A-2I) = \begin{pmatrix} -1 & 0 \\ 2 & 0 \end{pmatrix}$$

$-x_1 = 0$

$2x_1 = 0$

$\therefore \begin{pmatrix} 0 \\ k \end{pmatrix}$ 고유 vector

(2) $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix}$

$$\det(A-\lambda I) = \det\begin{pmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 1 & 3-\lambda \end{pmatrix}$$

$$= (1-\lambda)^2(3-\lambda) = 0$$

$\lambda = 1, 1, 3$

① $\lambda = 1$

$$A-I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

$x_2 + 2x_3 = 0$

$x_2 = -2x_3$

$\therefore \begin{pmatrix} k \\ 2s \\ -s \end{pmatrix}$ 고유 vector

② $\lambda = 3$

$$A-3I = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$-2x_1 = 0$

$-2x_2 = 0$

$x_2 = 0$

$\therefore \begin{pmatrix} 0 \\ 0 \\ k \end{pmatrix}$ 고유 vector

4.2절

1

(1) $\frac{dx}{dt} = 2x+3y$

$\frac{dy}{dt} = x+4y$

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$$

$$\det(A-\lambda I) = \det\begin{pmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{pmatrix}$$

$$= (2-\lambda)(4-\lambda) - 3 = 0$$

$$(\lambda-5)(\lambda-1) = 0$$

$$\lambda = 5, 1$$

① $\lambda = 5$

$$A-5I = \begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix}$$

$$-3x + 3y = 0$$

$$x = y$$

$$\therefore k_1\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

② $\lambda = 1$

$$A-\lambda I = \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix}$$

$$x+3y = 0$$

$$x = -3y$$

$$\therefore k_2\begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\therefore x(t) = k_1 e^{5t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + k_2 e^{t}\begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

2

(1) $\begin{cases} y_1' = 2y_1 + y_2 \\ y_2' = y_1 + 2y_2 \end{cases}$

$y_1(0) = -3, \quad y_2(0) = 5$

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$y' = Ay$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\det(A-\lambda I) = \det\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = 0$$

$$(\lambda-3)(\lambda-1) = 0$$

$$\lambda = 3, 1$$

① $\lambda = 3$

$$A - 3I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

$x_1 = x_2$

$\therefore k_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

② $\lambda = 1$

$$A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$x_1 = -x_2$

$\therefore k_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$\therefore y(t) = k_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + k_2 e^{t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} k_1 e^{3t} - k_2 e^{t} \\ k_1 e^{3t} + k_2 e^{t} \end{pmatrix}$$

$\therefore k_1 - k_2 = -3$

$k_1 + k_2 = 5$

$\therefore k_1 = 1, \quad k_2 = 4$

$$\therefore y(t) = \begin{pmatrix} e^{3t} - 4e^{t} \\ e^{3t} + 4e^{t} \end{pmatrix}$$

$y_1(t) = -4e^{t} + e^{3t}$

$y_2(t) = 4e^{t} + e^{3t}$

4. 3절

1

(1) $\begin{cases} y_1' + y_1 = t+1 \\ y_2' + 2y_2 = t^2 \end{cases}$ $\quad y' = Ay + g$

$$y' = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} y + \begin{pmatrix} t+1 \\ t^2 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \qquad -x_2 = 0$$

$\lambda = -1, -2 \quad \leftarrow \quad x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow y(t) = k_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-t} + k_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-2t}$

$y_p = u + vt + wt^2$ //

① $y_1' + y_1 = t+1$

$y_{p_1}(t) = At + B$

$A + At + B = t + 1$

$A = 1, \quad B = 0$

② $y_2' + 2y_2 = t^2$

$y_{p_2}(t) = At^2 + Bt + C$

$2At + B + 2At^2 + 2Bt + 2C = t^2$

$A = \frac{1}{2}, \quad B = -\frac{1}{2}, \quad C = \frac{1}{4}$

$$\therefore y(t) = k_1 e^{-t}\begin{pmatrix}1\\0\end{pmatrix} + k_2\begin{pmatrix}0\\1\end{pmatrix}e^{-2t} + \begin{pmatrix} t \\ \frac{1}{2}t^2 - \frac{1}{2}t + \frac{1}{4}\end{pmatrix}$$

#2

(3) $y_1' + y_2' - 13y_1 - 6y_2 = 0$ ⓐ

$y_1' - y_2' + 7y_1 + 2y_2 = 2e^t$ ⓑ

$y_1(0) = 0, \quad y_2(0) = -15$ ⓒ

$$y' = Ay + g \Rightarrow \begin{pmatrix} y_1' + y_2' \\ y_1' - y_2' \end{pmatrix} = \begin{pmatrix} 13y_1 + 6y_2 \\ -7y_1 - 2y_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 2e^t \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 13 & 6 \\ -7 & -2 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 2e^t \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\therefore \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} 13y_1 + 6y_2 + 0 \\ -7y_1 - 2y_2 + 2e^t \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2}(13y_1 + 6y_2) + \frac{1}{2}(-7y_1 - 2y_2 + 2e^t) \\ \frac{1}{2}(13y_1 + 6y_2) - \frac{1}{2}(-7y_1 - 2y_2 + 2e^t) \end{pmatrix}$$

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 3y_1 + 2y_2 + e^t \\ 10y_1 + 4y_2 - e^t \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 2 \\ 10 & 4 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} e^t \\ -e^t \end{pmatrix}$$

$$A = \begin{pmatrix} 3 & 2 \\ 10 & 4 \end{pmatrix}$$

$$\det(A - \lambda I) = \det\begin{pmatrix} 3-\lambda & 2 \\ 10 & 4-\lambda \end{pmatrix}$$

$$= (3-\lambda)(4-\lambda) - 20 = 0$$

$$\lambda^2 - 7\lambda + 12 - 20 = 0$$

$$(\lambda - 8)(\lambda + 1) = 0 \qquad \therefore \lambda = 8, -1$$

① $\lambda = 8$

$$A - 8I = \begin{pmatrix} -5 & 2 \\ 10 & -4 \end{pmatrix}$$

$$-5v_1 + 2v_2 = 0$$

$$5v_1 = 2v_2$$

$$\therefore \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

② $\lambda = -1$

$$A - (-1)I = \begin{pmatrix} 4 & 2 \\ 10 & 5 \end{pmatrix}$$

$$4v_1 + 2v_2 = 0$$

$$v_2 = -2v_1$$

$$\therefore \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$y_h(t) = c_1 \cdot \begin{pmatrix} 2 \\ 5 \end{pmatrix} e^{8t} + c_2 \begin{pmatrix} 2 \\ -1 \end{pmatrix} \cdot e^{-t}$$

$$y_p(t) = c \cdot e^t$$

$$c \cdot e^t = A\, c \cdot e^t + \begin{pmatrix} e^t \\ -e^t \end{pmatrix}$$

$$c = Ac + \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$(I - A)c = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} -2 & -2 \\ -10 & -3 \end{pmatrix}\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{cases} -2\alpha_1 - 2\alpha_2 = 1 \Rightarrow -10\alpha_1 - 10\alpha_2 = 5 \\ -10\alpha_1 - 3\alpha_2 = -1 \end{cases}$$

$$-7\alpha_2 = 6 \qquad \alpha_2 = -\frac{6}{7}$$

$$\alpha_1 = \frac{5}{14}$$

$$\therefore y_p(t) = \begin{pmatrix} \frac{5}{14} \\ -\frac{6}{7} \end{pmatrix} e^{t} \quad \text{④}$$

$$y(t) = c_1 \begin{pmatrix} 2 \\ 5 \end{pmatrix} e^{8t} + c_2 \begin{pmatrix} 2 \\ -1 \end{pmatrix} e^{-t} + \begin{pmatrix} 5/14 \\ -6/7 \end{pmatrix} e^{t} \quad //$$

$$\begin{cases} 2c_1 + 2c_2 + \frac{5}{14} = 0 \\ 5c_1 - c_2 - \frac{6}{7} = 0 \end{cases}$$

$$12c_1 - \frac{19}{14} = 0$$

$$c_1 = \frac{19}{14} \cdot \frac{1}{12} = \frac{19}{168}$$

$$\frac{95}{168} - c_2 = \frac{144}{168}$$

$$c_2 = \frac{-49}{168} = -\frac{7}{24} \quad //$$

$$\therefore y(t) = 19/168 \begin{pmatrix} 2 \\ 5 \end{pmatrix} e^{8t} + (-7/24) \begin{pmatrix} 2 \\ -1 \end{pmatrix} e^{-t} + \begin{pmatrix} 5/14 \\ -6/7 \end{pmatrix} e^{t} \quad //$$

추가 문제

(1)
$$\begin{cases} y_1' = 10y_1 - 10y_2 - 4y_3 \\ y_2' = -10y_1 + y_2 - 14y_3 \\ y_3' = -4y_1 - 14y_2 - 2y_3 \end{cases}$$

$$y' = Ay$$

$$A = \begin{pmatrix} 10 & -10 & -4 \\ -10 & 1 & -14 \\ -4 & -14 & -2 \end{pmatrix}$$

$$\det(A - \lambda I) = \det \begin{pmatrix} 10-\lambda & -10 & -4 \\ -10 & 1-\lambda & -14 \\ -4 & -14 & -2-\lambda \end{pmatrix}$$

$$= 10-\lambda\left((1-\lambda)(-2-\lambda) - 196\right) + 10 \cdot \left(10(2+\lambda) - 56\right) - 4\left(140 + 4(1-\lambda)\right)$$

$$= (10-\lambda)(\lambda^2 + \lambda - 198) + 10(10\lambda - 36) - 4(-4\lambda + 144)$$

$$= 10\lambda^2 + 10\lambda - 1980 - \lambda^3 - \lambda^2 + 198\lambda + 100\lambda - 360 + 16\lambda - 576$$

$= -\lambda^3 + 9\lambda^2 + 324\lambda - 2916$

$\lambda = 18, -18, 9$

① $\lambda = 18$ ② $\lambda = -18$ ③ $\lambda = 9$

$$A - 18I = \begin{pmatrix} -8 & -10 & -4 \\ -10 & -17 & -14 \\ -4 & -14 & -20 \end{pmatrix} \qquad V_2 = \begin{pmatrix} 1/2 \\ 1 \\ 1 \end{pmatrix} \qquad V_3 = \begin{pmatrix} -1 \\ -1/2 \\ 1 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

$$y(t) = c_1 \begin{pmatrix} 1/2 \\ 1 \\ 1 \end{pmatrix} e^{-18t} + c_2 \begin{pmatrix} -1 \\ -1/2 \\ 1 \end{pmatrix} e^{9t} + c_3 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} e^{18t}$$

(2) $y_1' = y_1 - y_2 + 0$

$y_2' = y_1 + 3y_2 - e^{-2t}$ ⑪

$y' = Ay + g$

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$$

$\lambda^2 - 4\lambda + 3 + 1$

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & -1 \\ 1 & 3-\lambda \end{pmatrix} = 0$$

$\lambda = 2, 2$

$\lambda = 2$

$$A - 2I = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$$

$(A - 2I)\,u = V$

$$u = \begin{pmatrix} 1 - u_2 \\ u_2 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$y_h(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t} + c_2 \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{2t}$$

$y_p = a \cdot e^{-2t}$

$y_p' = -2a \cdot e^{-2t}$

$\Rightarrow \quad -2a \cdot e^{-2t} = A \cdot a e^{-2t} + \begin{pmatrix} 0 \\ -e^{-2t} \end{pmatrix}$

$-2a = Aa + \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

$(+2+A)\,a = \begin{pmatrix} 0 \\ +1 \end{pmatrix}$

$\begin{pmatrix} 3 & -1 \\ 1 & 5 \end{pmatrix} a = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$3a_1 - a_2 = 0$

$a_1 + 5a_2 = 1$

$16a_1 = 1$

$a_1 = 1/16$

$a_2 = 3/16$

$\therefore \quad y(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t} + c_2 \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{2t} + \begin{pmatrix} 1/16 \\ 3/16 \end{pmatrix} e^{-2t}$